



Regression Analysis



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Covariance of Two Numerical Variables

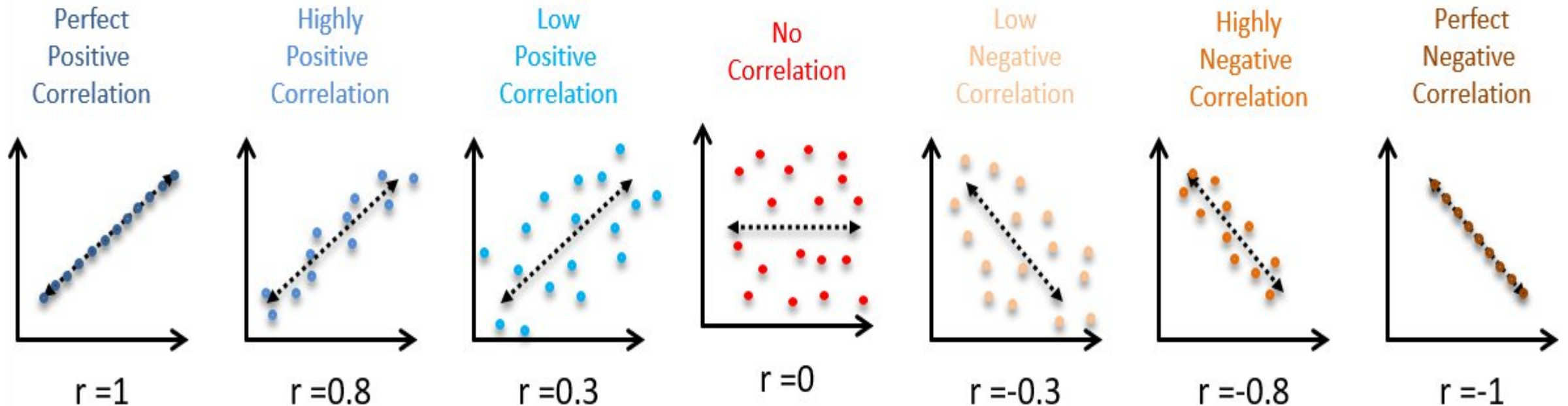
Covariance will measure joint variation of two numerical variables. This is a measure of the relationship between two variables.

$$COV(X,Y) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Correlation of Two Numerical Variables

Correlation is also a measurement of the relationship between two numerical variables. It is lying between -1 and +1.

$$CORR(X,Y) = \frac{COV(X,Y)}{\sqrt{V(X)V(Y)}}$$



Correlation VS Causation



A correlation between variables, however, does not automatically mean that the change in one variable is the cause of the change in the values of the other variable. Causation indicates that one event is the result of the occurrence of the other event; i.e. there is a causal relationship between the two events.

Linear Regression

Linear regression is perhaps one of the most well known and well understood algorithms in statistics and machine learning.

Linear regression is a linear model, e.g. a model that assumes a linear relationship between the input variables (x) and the single output variable (y). More specifically, that y can be calculated from a linear combination of the input variables (x).

When there is a single input variable (x), the method is referred to as **Simple Linear Regression**. When there are multiple input variables, literature from statistics often refers to the method as **Multiple Linear Regression**.

Different techniques can be used to prepare or train the linear regression equation from data, the most common of which is called **Ordinary Least Squares**. It is common to therefore refer to a model prepared this way as Ordinary Least Squares Linear Regression or just Least Squares Regression.

Simple Linear Regression

Assume that we want to see the relationship between the height and the weight of Sri Lankan university students. Our objective is to fit a model to predict the weight using height of the students.

The equation for this model for the population is as follows,

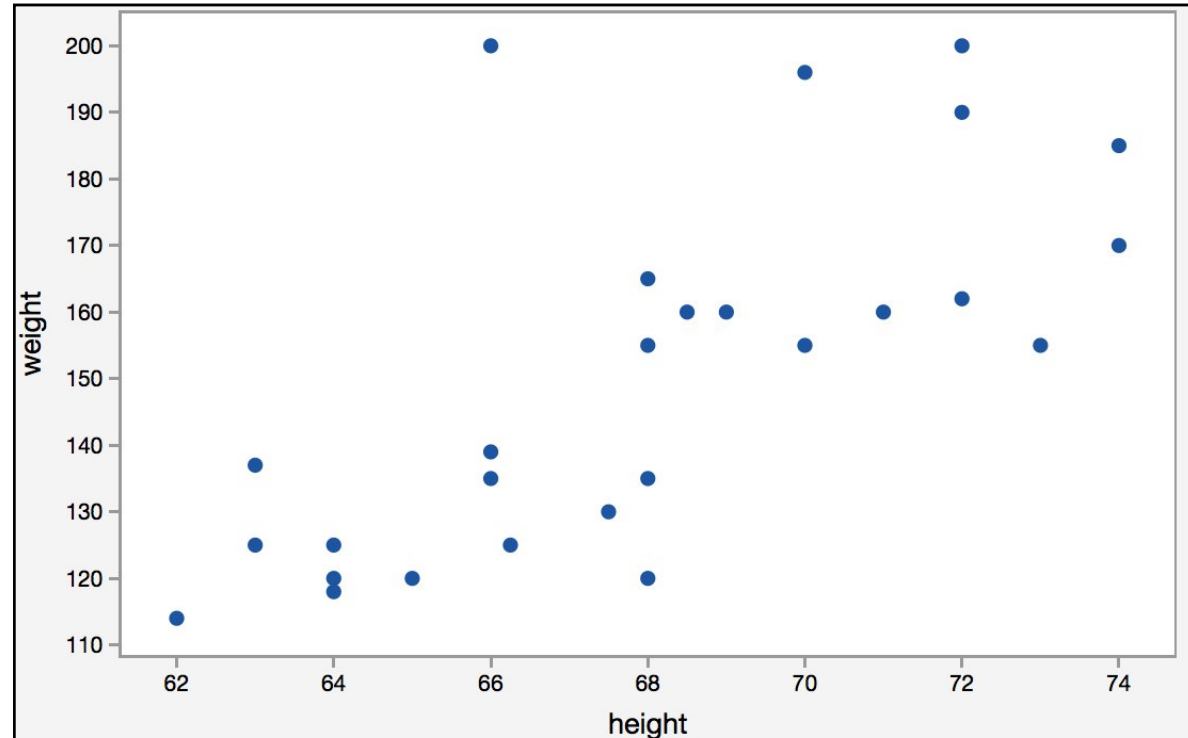
$$y = \beta_0 + \beta_1 x + \varepsilon$$

Since the entire population cannot be accessed, a sample will be taken, and the model will be created.

Let's plot the sample data.

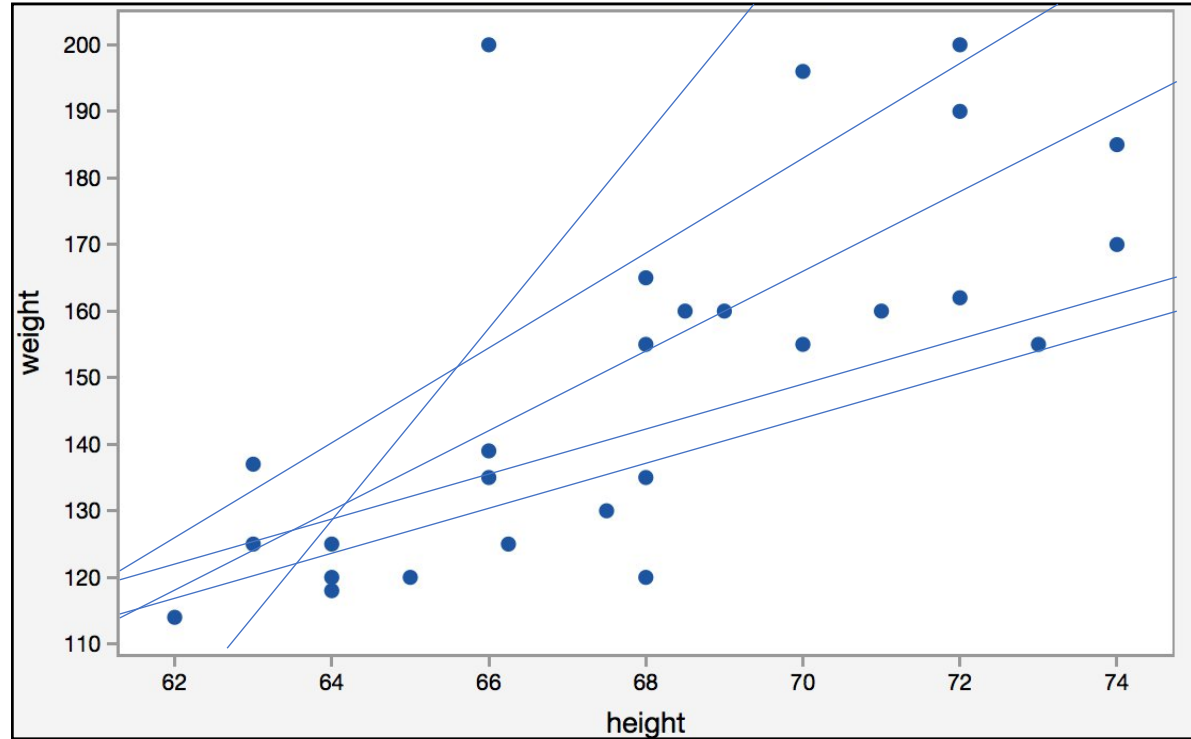


Simple Linear Regression



Simple linear regression is useful for finding relationship between two variables. One is predictor or independent variable and other is response or dependent variable which is a quantitative variable. For example, relationship between height and weight.

Simple Linear Regression



Infinitely many number of plots can be created.

Simple Linear Regression

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

The values $\hat{\beta}_0$ and $\hat{\beta}_1$ must be chosen so that they minimize the error. If sum of squared error is taken as a metric to evaluate the model, then goal to obtain a line that best estimated model which reduces the error.

$$\text{Sum of Squares of Error (SSE)} = \sum_{i=1}^n (\text{Actual Output} - \text{Predicted Output})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

This is called as Residual Sum of Squares (RSS) as well. By minimizing this SSE, we can obtain following parameter estimations. Significance of these parameters can also be measured using **Hypothesis Testing**.

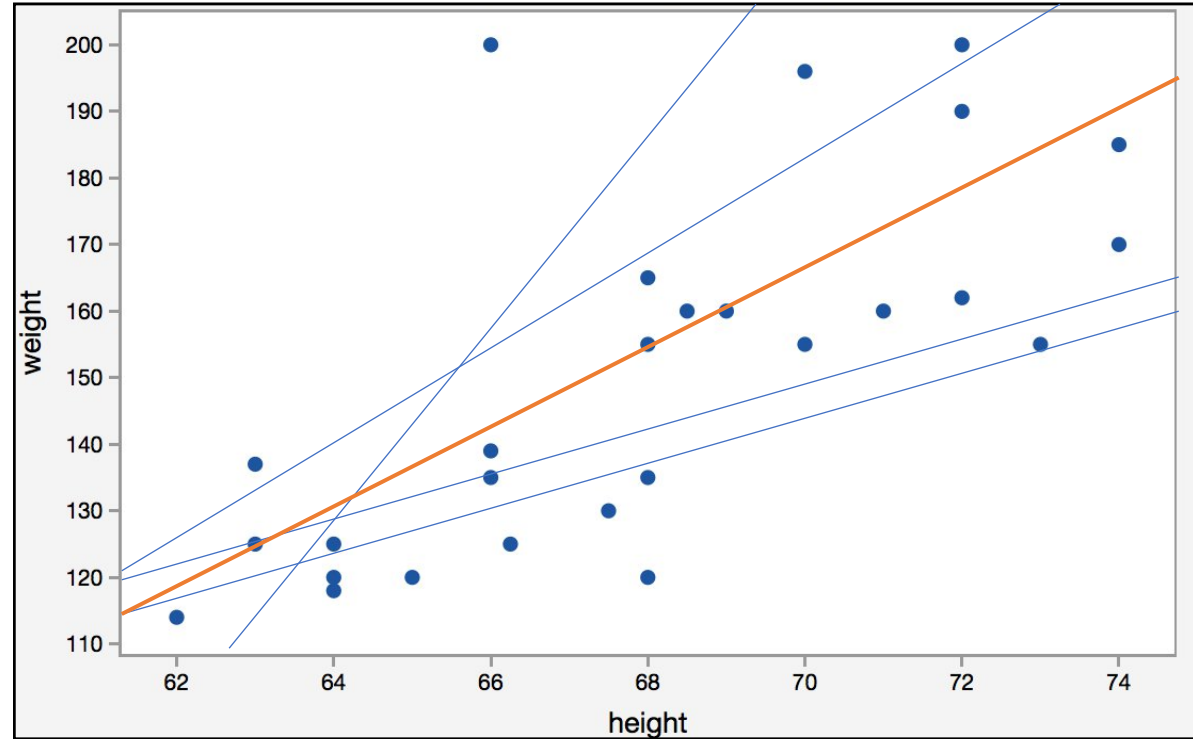
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Then the estimated model is,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Simple Linear Regression



Best Fit Line

The core idea is to obtain a line that best fits the data. The best fit line is the one for which total prediction error (all data points) are as small as possible.

Multiple Linear Regression

The equation for this model is as follows,

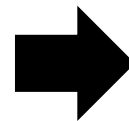
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_k x_k + \varepsilon$$

Consider here we have k variables. By minimizing this SSE, we can obtain the parameter estimations in here as well. These estimated parameters can be represented as a vector. The parameter vector $\underline{\hat{\beta}}$ can be obtained through,

$$\underline{\hat{\beta}} = (X^T X)^{-1} X^T Y$$

Here,

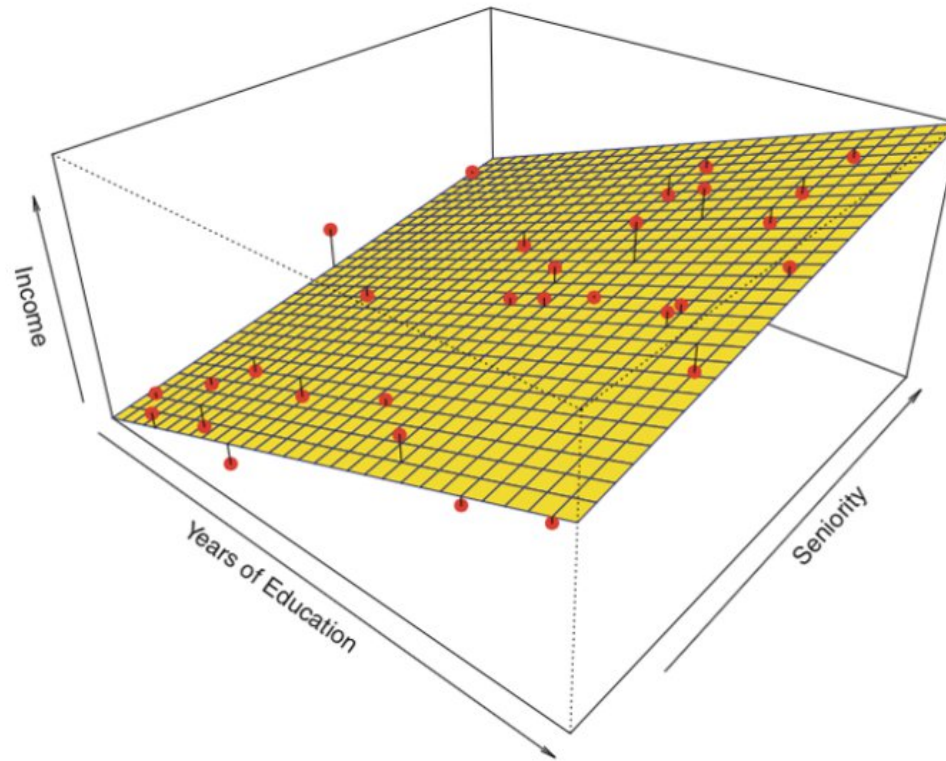
X_1	X_2	...	X_k
X_{11}	X_{12}	...	X_{1k}
X_{21}	X_{22}	...	X_{2k}
...	...	...	...
X_{n1}	X_{n2}	...	X_{nk}



$$\mathbf{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}$$

Multiple Linear Regression

How this model is visualized. Consider a 2 variables with one response example. Here the **Income** is the response variable and **Seniority** and the **Years Of Education** are the independent variables.



Higher dimensions cannot be visualized easily. There are several advanced techniques for visualize them.

Qualitative Predictors (Categorical Independent Variables)

Categorical variables cannot be added to the model as the numerical variables.

Ex- University Year Variable (First, Second, Third, Fourth)

First	Second	Third	Fourth
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

These are called as **Dummy Variables**.

Qualitative Predictors (Categorical Independent Variables)

Since one variable can be represented as other variable's 0 case, one variable can be removed. That is called as the **reference level**.

Consider **First** as the reference level.

First	Second	Third	Fourth
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

This is called as **Dummy Trapping**.

Qualitative Predictors (Categorical Independent Variables)

Since one variable can be represented as other variable's 0 case, one variable can be removed. That is called as the **reference level**.

Second	Third	Fourth
0	0	0
1	0	0
0	1	0
0	0	1

This is called as **Dummy Trapping**.

Dummy Variables

To represent categorical variables in a linear regression model, we use dummy variables. Consider following example.

Ex:- Gender

Gender	Dummy Variable (G)
Male	1
Female	0

Ex- Temperature (High, Medium, Low)

Temperature	Dummy Variable (T1)	Dummy Variable (T2)
High	1	0
Medium	0	1
Low	0	0

Ex- Colour (Red, Green, Yellow, Blue)

Colour	Dummy Variable (C1)	Dummy Variable (C2)	Dummy Variable (C3)
Red	1	0	0
Green	0	1	0
Yellow	0	0	1
Blue	0	0	0

Dummy Variables

Now assume that the model to be fitted for a numerical variable (X) and the above discussed categorical variable **Colour**.

Colour	Dummy Variable (C1)	Dummy Variable (C2)	Dummy Variable (C3)
Red	1	0	0
Green	0	1	0
Yellow	0	0	1
Blue	0	0	0

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 c_1 + \hat{\beta}_3 c_2 + \hat{\beta}_4 c_3$$

R Squared Value (Coefficient of Determination)

$$\text{Total Sum of Squares (TSS)} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\text{Sum of Squares of Error (SSE)} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{Coefficient of Determination} = R^2 = \frac{TSS - SSE}{TSS}$$

This R Squared Value is explaining the fraction of variation explained by the estimated model. In simple words how much of the data captured by this model.

- If the R Squared value is close to 1, it means the model is well fitted.
- If the R Squared value is close to 0, it means the model is underfitted.

For the Simple Linear Regression case $\text{Coefficient of Determination} = R^2 = (\text{Correlation})^2$

Hypothesis Testing for Simple Linear Regression

For each sample of the population, different estimates can be obtained. So these estimates are random variables.

$$V(\hat{\beta}_0) = \frac{SSE}{n-2} \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

$$V(\hat{\beta}_1) = \frac{SSE}{n-2} \left[\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

Hypothesis Testing for Simple Linear Regression

Following hypothesis can be tested.

H_0 : *There is no relationship between X and Y* ($\beta_1 = 0$)

H_1 : *There is a relationship between X and Y* ($\beta_1 \neq 0$)

Here the test statistic is,

$$t_{cal} = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \sim t_{n-2}$$

Hypothesis Testing for Multiple Linear Regression

Following hypothesis can be tested.

$$H_0: \beta_1 = \beta_2 = \beta_3 = \dots = \beta_k = 0$$

$$H_1: \text{At least one } \beta_j \text{ is non zero for } j = 1, 2, \dots, k$$

ANOVA table here is,

Source of variation	Degrees of freedom	Mean square	F
Regression	k	$MSR = \frac{TSS - SSE}{k}$	$F_{stat} = \frac{MSR}{MSE}$
Error	n-k-1	$MSE = \frac{SSE}{n-k-1}$	
Total	n-k-1+(k)=n-1		

Linear Regression Assumptions

There are mainly four assumptions associated with a linear regression model:

- **Linearity:** The relationship between X and the mean of Y is linear.
- **Multicollinearity:** Correlation among independent variables.
- **Homoscedasticity:** Homogeneity of variance of residuals.
- **Independence:** Observations are independent of each other.
- **Normality:** Residuals are normally distributed.

Linearity

If you fit a linear model to a non-linear, non-additive data set, the regression algorithm would fail to capture the trend mathematically, thus resulting in an inefficient model. Also, this will result in erroneous predictions on an unseen data set.

- How to check:
 - Look for residual vs fitted value plots. It must not contain any regular pattern.
 - Scatter plots between the response variable and each independent variable.
 - Partial linear regression models between the response variable and each independent variable.
- Also, you can include polynomial terms (X , X^2 , X^3) in your model to capture the non-linear effect.

Multicollinearity

Multicollinearity means independent variables in a model are correlated.

- Multicollinearity among independent variables can reduce the performance of the model.
- Multicollinearity can be a problem in multiple regression because the input variables are all influencing each other.
- Therefore, they are not actually independent, and it is difficult to test how much the combination of the independent variables affects the dependent variable or outcome.
- Hence we need **Variance Inflation Factor (VIF)** as it is a tool to help measure the degree of multicollinearity.

$$VIF_i = \frac{1}{1 - R_i^2}$$

Multicollinearity

We will be calculating the VIF of each feature individually, for that we need to calculate R^2 the equation for regression for each variable individually with the response variable.

VIF value will always be greater than 1. Here are some rules for VIF,

- 1 = not correlated.
- Between 1 and 5 = moderately correlated.
- Greater than 5 = highly correlated.

Homoscedasticity

Residuals must have the constant variance.

- The presence of non-constant variance in the error terms results in heteroskedasticity.
- Generally, non-constant variance arises in presence of outliers or extreme leverage values.
- How to check:
 - You can look at residual vs fitted values plot.
 - If heteroskedasticity exists, the plot would exhibit a funnel shape pattern (shown in next section).
 - Also, you can use Breusch-Pagan / Cook - Weisberg test or White general test to detect this phenomenon.

Independence of Observations

There should be no correlation between the residual (error) terms. Absence of this phenomenon is known as Autocorrelation.

- The presence of correlation in error terms drastically reduces model's accuracy.
- This usually occurs in time series models where the next instant is dependent on previous instant.
- If the error terms are correlated, the estimated standard errors tend to underestimate the true standard error.
- How to check:
 - Residuals Vs Fitted Value plot. It must not contain any regular pattern.
 - Look for Durbin – Watson (DW) statistic.

Independence of Observations

- Durbin – Watson (DW) statistic.

If e_t is the residual given by $e_t = \rho e_{t-1} + \nu_t$, the Durbin-Watson test statistic is

$$d = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2},$$

- It must lie between 0 and 4.
- If DW = 2, implies no autocorrelation,
- $0 < DW < 2$ implies positive autocorrelation
- $2 < DW < 4$ indicates negative autocorrelation.

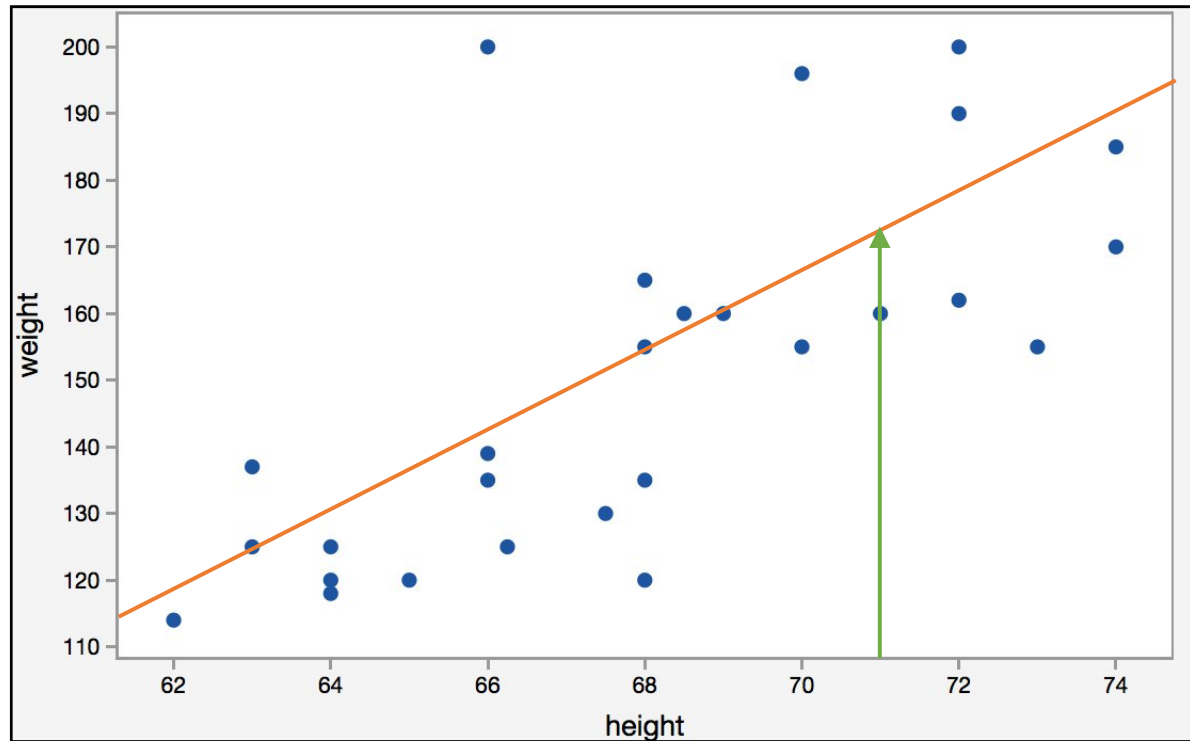
Normality of Residuals

Residuals must be normally distributed.

- How to check:
 - Q-Q plot.
 - Shapiro-Wilk test

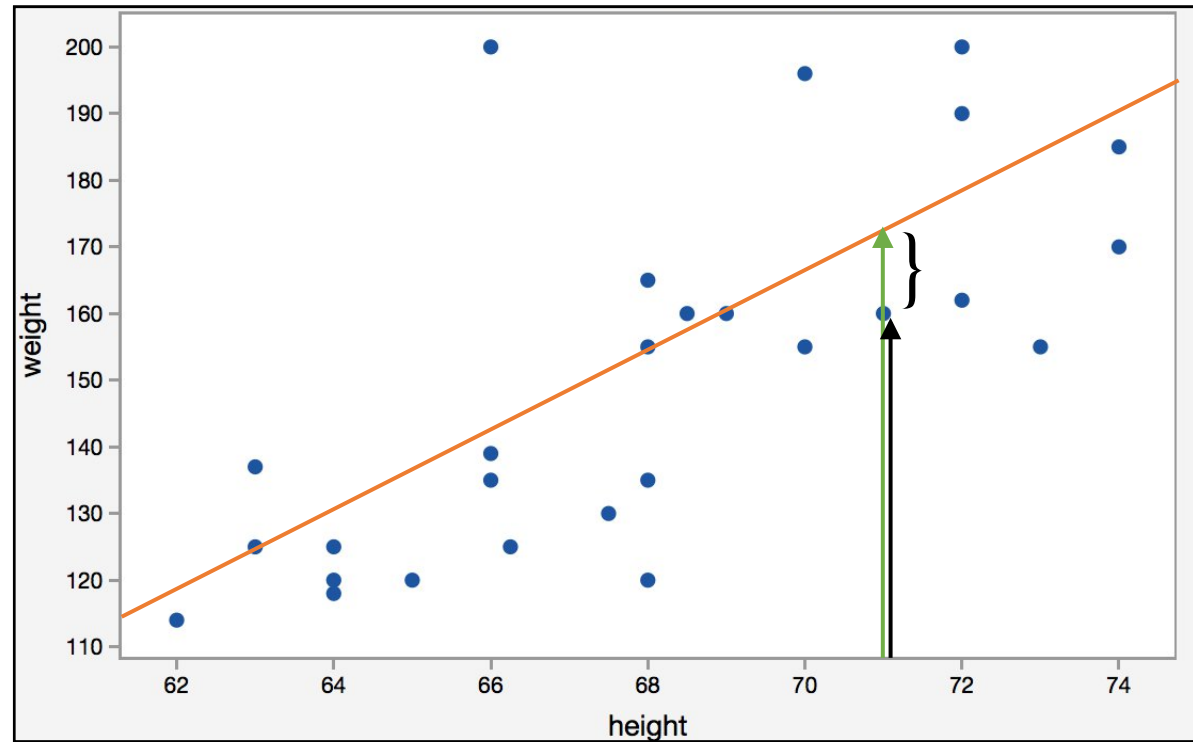
Predictions

After fitting the model, the goal is to predict the response using independent data.



Prediction Error

Always there is a prediction error.



Model Evaluation – Validation Set Approach

Split the dataset into two sets,

- Training dataset (Generally 80 of the data But it can be changed)
- Testing dataset (Rest of the data)

Train the model using the training dataset and then check the accuracy of the model using the testing dataset. MSE of a regression model will be calculated and the model will be evaluated.

$$\text{Mean Squared Error} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$